

KOSLAN, Russian Federation

WMO: 226950

Lat: 63.474N

Lon: 48.845E

Elev: 402

StdP: 14.48

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-21.9	-32.5	0.9	-27.0	-27.0	1.2	-21.8	18.3	13.7	16.5	11.5	3.5	80	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	82.2	65.5	77.8	63.3	74.1	61.8	67.6	77.6	65.7	74.4	63.6	71.5	6.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	91.2	71.1	62.2	85.0	70.0	60.2	79.0	67.9	32.3	77.5	30.8	74.8	29.1	71.4	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.5	15.3	13.8	DB	-32.0	86.0	8.6	3.8	-38.2	88.7	-43.2	91.0	-48.1	93.1	-54.3	95.9	(4)
(5)			WB	-32.2	70.3	8.5	3.0	-38.4	72.5	-43.4	74.3	-48.2	76.0	-54.4	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.4	5.0	11.1	20.4	32.8	44.6	55.5	61.6	55.8	45.7	34.2	21.4	11.9		
	DBStd	21.41	14.60	14.10	10.14	8.47	10.45	8.67	7.63	7.51	6.64	7.14	11.78	15.62		
	HDD50	6917	1395	1090	919	516	239	39	6	26	158	491	858	1180		
	HDD65	11609	1860	1510	1384	965	636	308	158	302	578	955	1308	1645		
	CDD50	877	0	0	0	1	72	203	364	205	30	2	0	0		
	CDD65	92	0	0	0	0	4	22	51	15	0	0	0	0		
	CDH74	876	0	0	0	0	55	223	485	112	1	0	0	0		
	CDH80	213	0	0	0	0	9	52	123	29	0	0	0	0		
	WSAvg	6.8	6.4	6.9	7.2	7.7	7.6	7.2	6.2	6.1	6.4	7.0	6.9	6.4		
	PrecAvg	23.8	1.5	1.3	1.2	1.5	1.7	2.6	2.4	3.1	2.3	2.6	2.0	1.7		
(15) Precipitation	PrecMax	27.6	2.1	2.9	2.0	2.4	3.4	5.1	3.6	5.6	4.5	4.7	4.1	2.6		
	PrecMin	19.1	0.7	0.5	0.1	0.6	0.5	0.7	0.5	1.5	0.9	0.9	0.4	0.7		
	PrecStd	2.7	0.4	0.6	0.6	0.6	0.7	1.3	0.9	1.1	0.8	1.1	0.9	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.8	34.9	44.4	62.8	80.8	85.7	87.9	83.2	69.5	54.6	42.5	33.9		
		MCWB	32.0	33.1	36.2	48.3	59.9	66.8	68.2	67.1	59.0	51.0	41.2	33.1		
	2%	DB	30.4	32.5	38.5	53.6	73.0	80.1	83.5	76.2	63.4	49.3	37.8	32.8		
		MCWB	29.9	31.8	33.3	43.0	56.5	63.7	66.6	64.3	55.8	47.5	37.4	32.2		
	5%	DB	27.8	30.7	35.5	48.1	67.8	75.6	79.2	71.7	59.3	46.1	35.2	31.8		
		MCWB	27.2	29.9	31.8	39.2	53.4	61.0	64.6	62.3	53.7	44.4	34.8	31.4		
	10%	DB	24.0	28.4	33.4	44.5	62.6	71.0	75.6	67.7	55.9	43.4	33.4	30.3		
		MCWB	23.3	27.6	30.9	37.8	51.2	59.3	63.0	60.3	51.9	42.3	32.9	29.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.3	33.7	37.2	49.4	62.3	68.7	71.3	69.5	59.9	52.1	42.0	33.3		
		MCDB	32.5	34.3	42.5	61.1	76.1	80.2	82.6	81.3	67.7	53.7	42.5	33.5		
	2%	WB	29.9	31.9	34.4	44.2	59.0	66.1	68.4	66.2	56.9	47.8	37.3	32.3		
		MCDB	30.3	32.3	37.1	52.0	71.4	76.3	79.1	73.8	61.8	49.1	37.7	32.5		
	5%	WB	27.4	29.9	32.8	40.6	55.2	63.5	66.6	63.8	54.5	44.6	35.0	31.4		
		MCDB	27.9	30.6	34.8	46.3	66.1	72.5	75.9	69.6	58.4	45.8	35.2	31.7		
	10%	WB	23.3	27.6	31.3	38.2	51.7	60.9	64.6	61.2	52.2	42.4	33.0	29.6		
		MCDB	23.8	28.3	33.3	42.9	61.0	69.4	73.0	66.6	55.3	43.5	33.3	30.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	11.0	14.5	14.4	17.0	16.9	17.4	14.6	11.6	6.8	7.4	9.6		
		MCDBR	9.4	7.1	14.1	21.2	24.3	23.8	23.2	20.1	19.0	10.5	7.1	6.1		
		MCWBR	9.4	6.5	10.3	13.2	13.0	11.0	9.9	9.9	11.9	8.5	7.1	6.3		
	5% WB	MCDBR	9.6	6.7	10.3	17.7	22.6	20.6	20.4	18.6	16.6	9.4	7.0	6.0		
		MCWBR	9.6	6.3	8.2	12.2	13.2	11.5	10.2	10.4	11.1	8.2	7.1	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.232	0.282	0.314	0.348	0.347	0.352	0.398	0.366	0.331	0.304	0.234	0.218		
	taud		1.948	2.060	2.090	2.132	2.415	2.474	2.345	2.415	2.502	2.351	2.022	1.925		
	Ebn at Noon		110	197	241	260	275	276	257	256	244	197	109	49		
	Edh at Noon		10	22	32	38	32	30	34	29	22	16	9	5		
(44) All-Sky Solar Radiation	RadAvg		61	269	702	1259	1565	1664	1661	1133	602	233	64	25		
	RadStd		7	51	102	124	112	168	140	98	67	31	14	3		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page